Distill: 10-minute pitch guide

What to say, what to show, and how to answer questions.

The one sentence

Ericsson deletes recordings when storage is full, and the knowledge disappears with them. Distill keeps the knowledge (every slide, every spoken word, who said it and when) and drops the weight: 1.45 GB of Ericsson's files became 25 MB, and we can prove nothing important was lost.

Key numbers (memorise these)

What	Number	Say it like this
All 12 files	1.45 GB → 25 MB	98.3% less storage, official documents kept as originals
Meeting, 155 min	1.14 GB → 7.5 MB	99.4% smaller, still every slide, word and speaker
GraalVM talk, 16 min	210 MB → 0.62 MB	99.7% smaller
Slide text kept	95.8% · 99.7%	Meeting · GraalVM, checked at 20 moments
Pictures match	90% · 100%	Pixel comparison, no OCR, so the tool doesn't grade itself
OCR accuracy	95.1%	Checked against the true text of 36 official slides
SD-WAN Rev A	3.30 MB → 0.20 MB	Only 7 changed slides stored, 18 shared with Rev B
Employee leaving	86%	of Gonzalo's drive can be freed, nothing lost

The 10-minute plan

Time	Show	Say (short)
0:00	Slides 1–2	Ericsson's problem: recordings fill storage, get deleted, knowledge is lost; copies everywhere.
1:00	Slide 3	Our result: 1.45 GB → 25 MB, 98.3% less, on Ericsson's own sample files.
1:45	Slide 4	Find → Distill → Link → Verify → Use. Four offline AI models, the rest simple algorithms.
2:30	**Live demo:** GraalVM file page	Move the slider from Level 1 to 5; open 'What is stored', show transcript and slides; download Level 4 zip.
4:00	Slides 5–7	The meeting at five levels; Level 4 is our default: 1.14 GB → 7.5 MB. All videos 99.4%.
5:00	Slide 8 + **demo:** Duplicates	Official copy kept, copies become references, Rev A stored as changes only. Click Rebuild.
6:15	Slide 9 + **demo:** evidence	Proof: 95.8% slide text, 90% pictures match (no OCR), 95.1% OCR accuracy vs official slides.
7:15	Slide 10 + **demo:** Employee leaving	Gonzalo leaves: preview first; 86% freed safely; quarantine, never delete.
8:15	**Demo:** Search	Search 'humanitarian': exact slide + sentence + time, from the 25 MB archive only.
9:00	Slides 11–12	All offline on a laptop. Beyond the brief + next steps. Close with the tagline.

How it works, in simple words

Use these if a judge asks how something works. One idea per box.

1. Videos: keep the information, drop the moving picture

Whisper turns speech into text with a time for every word. **pyannote + WeSpeaker** work out who spoke. **OpenCV** notices every slide change (ignoring the webcam corner) and keeps one picture per slide, taken just before it changes so all bullets are visible. **RapidOCR** reads each slide. Audio is compressed with Opus, pictures with WebP, text with gzip.

2. Five levels: Ericsson chooses how much to keep

Level 1 original video · Level 2 smaller video · Level 3 audio + slides + text · **Level 4 slides + text (our default)** · Level 5 text + links to the official deck. Each level is just the sum of the files it keeps.

3. Documents: keep one, reference the rest

Every page gets two fingerprints: SHA-256 of its text (footers and page numbers removed) answers 'same slide?', pHash of its picture answers 'looks the same?'. The official copy stays as the original file; exact copies become a reference (0 bytes); an older revision keeps only its changed slides. We read Ericsson's document control, so final2.pptx is correctly Rev A from 2019.

4. Link: connect video slides to the official decks

Each slide seen in a video is compared with all deck pages: 75% word overlap + 25% picture similarity. A score of 55% or more creates a link, e.g. 'meeting at 17:35 = Ericsson Response PDF, page 2'. 36 of the meeting's slides are linked.

5. Proof: nothing important was lost

We pause the original video at 20 evenly spaced moments (length × (i + 0.5) ÷ 20) and compare the screen with the slide we kept. Text moments are checked by text, still pictures by how they look, moving video is reported separately. Names that are always on screen are ignored. Two independent extra checks stop OCR grading itself: a pixel comparison with no OCR, and OCR checked against the true text of official slides.

6. Search and safe clean-up

Every 90 seconds of video (slide text + spoken words) and every document page becomes a search piece, findable by exact words and by meaning. Search runs only on the compressed archive. When someone leaves, a file is removed only if an official copy exists; partly unique decks keep their unique slides; everything is previewed, quarantined (restorable) and logged.

Technology at a glance

AI models (4, all offline)	Algorithms and tools
Whisper base: speech to text	FFmpeg, Opus, WebP, gzip: compression
pyannote + WeSpeaker: who spoke when	SHA-256, pHash, dHash: fingerprints
RapidOCR: reads slide text	PyMuPDF, python-pptx, LibreOffice: documents
all-MiniLM-L6-v2: search by meaning	SQLite + FTS5: database and keyword search
	FastAPI + Next.js: backend and interface

Likely questions, short answers

Answer in one or two sentences, then offer to show it in the app.

Question	Short answer
Everyone can compress 99%. What's different?	Anyone can throw data away. We keep what matters, link every slide to its official deck and every sentence to its moment, and we measure that 95–100% of what was on screen survived.
What if the missing 5% is confidential?	It isn't deleted content: it's mostly tiny text OCR couldn't read, and the kept slide picture still contains it. Nothing is reduced automatically; for confidential or legal material Ericsson keeps Level 1.
Isn't OCR grading itself?	That's why we added two independent checks: a pixel comparison with no OCR (90% / 100%) and OCR checked against the true text of 36 official slides (95.1%).
Why not just keep the video at lower quality?	That's Level 2 and it's available. But it's still ~165 MB for the meeting; slides + text is 7.5 MB and is searchable.
How long does processing take?	About half the video's length on a laptop CPU: the 155-minute meeting took about 72 minutes. A GPU or a Teams transcript makes it much faster.
Two different talks using the same slides?	Not treated as duplicates: video duplicates must also have at least 80% similar transcripts.
How do you know which deck version is newer?	We read Ericsson's document control in the footer (e.g. 'Uen | A | 2019-10-11'), not the file name. That's how we found final2.pptx is actually Rev A.
Could the clean-up delete something important?	No. A file is only 'safe' if an official copy exists; we never trust another person's drive; we preview first, quarantine instead of deleting, and log every action.
Does any data leave the machine?	No. All models run offline on the laptop; personal data is removed before any AI export.
Does it scale to all of Ericsson?	The pipeline is per file, so it runs in parallel. For millions of slides the next step is MinHash for near-duplicate search, plus sensitivity labels and legal-hold rules.

Before you start

- ☐ Backend and frontend running (start_backend.bat, start_frontend.bat); open localhost:3000, never port 8000.
- ☐ Everything already processed. Don't process the big video live; a PDF or the Jury video is fine.
- ☐ Have a short screen recording of the demo as a backup.
- ☐ Sizes in the app use 1 GB = 1024 MB, so the meeting is 1.14 GB (not 1.22 GB). Use the app's numbers.
- ☐ Know your weak spots and say them first: speaker detection needed merging, English-only meaning search, processing time.